

A Multi-Stage Machine Learning Pipeline for Automated Bowel Preparation Scale Assessment in Colonoscopy Videos

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Abstract—Accurate assessment of intestinal cleanliness is essential for effective colonoscopy, but manual scoring with the Boston Bowel Preparation Scale (BBPS) remains subjective and labor-intensive. In this study, we propose a novel, fully automated multi-stage pipeline for objective BBPS-based bowel preparation assessment in colonoscopy videos. Our approach utilizes deep image classifiers, enhanced by a hierarchical sequence of binary classification tasks, to assign precise frame-level BBPS scores. From the resulting sequence of scores, we extract statistical and temporal features to train machine learning regression models for video-level BBPS prediction. Extensive experiments demonstrate that our pipeline achieves robust, reproducible, and granular video-level bowel cleanliness assessment, outperforming standard multi-class models at the frame level and offering a scalable tool for large-scale endoscopic quality analysis.

Index Terms—Image classification, Machine learning, Colonoscopy

I. INTRODUCTION

Colonoscopy is a pivotal procedure for visualizing the colonic mucosa to detect and manage abnormalities, but its efficacy depends heavily on the bowel preparation quality. Inadequate cleansing can obscure lesions, reducing diagnostic accuracy and forcing repeat procedures. Therefore, a standardized assessment of bowel cleanliness is essential to ensure optimal outcomes.

Introduced by Lai et al. in 2009 [1], the BBPS is a validated tool designed to standardize the evaluation of bowel cleanliness during colonoscopy, which classifies colonoscopy images into five categories: class 0 (excellent, completely clean), class 1 (good, minor residue), class 2 (fair, moderate residue), class 3 (poor, significant residue), and class 4 (redundant, not evaluable due to inadequate visibility). The BBPS divides the colon into three segments: the right colon (including the cecum and ascending colon), the transverse colon (including hepatic and splenic flexures), and the left colon (including the descending colon, sigmoid colon, and rectum). Each segment is scored on a scale from 0 to 3, with higher scores indicating better preparation quality; thus, the total score ranges from 0 to 9. Despite the BBPS's widespread adoption, challenges persist. Studies have shown that even with standardized scales, there can be significant variability between observers, affecting clinical outcomes and research data reliability. Proper use of the BBPS necessitates training; without adequate instruction,

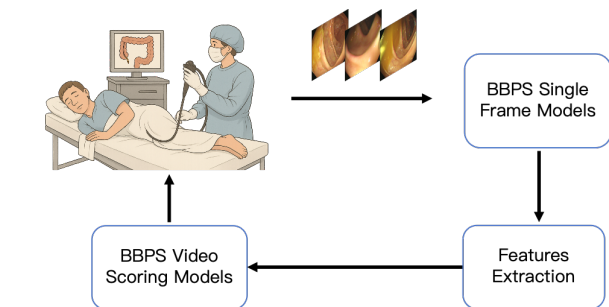


Fig. 1: Pipeline of Automated Bowel Preparation Scale

the reliability of the scale diminishes, leading to potential misclassification of bowel cleanliness. In busy clinical settings, thorough assessment using the BBPS can be time-consuming, potentially leading to hurried evaluations and inaccuracies.

This paper develops a comprehensive, multi-stage pipeline to automate BBPS-based bowel cleanliness assessment in colonoscopy videos, advancing beyond conventional approaches in several key aspects. Figure 1 shows the pipeline for automated bowel preparation scale assessment. We first employ transfer learning to train image classifiers that predict frame-level BBPS scores. At the core of our method, we introduce a hierarchical sequence model strategy, where the BBPS classification is reformulated from a standard single-model 5-class prediction into a structured sequence of binary (2-class) classifiers. This hierarchical design enables the model to progressively distinguish frame quality with finer granularity, significantly improving classification accuracy compared to direct multi-class prediction. The resulting frame-level scores are then aggregated across the entire video, from which we extract a broad spectrum of statistical, consistency, and trend-based features that capture both local and global quality dynamics. These features serve as input to a suite of machine learning models for video-level BBPS scoring. Our pipeline achieves scalable, reproducible, and fine-grained assessment of bowel preparation quality, and our hierarchical classification approach provides a novel and effective alternative to conventional single-model classification frameworks.

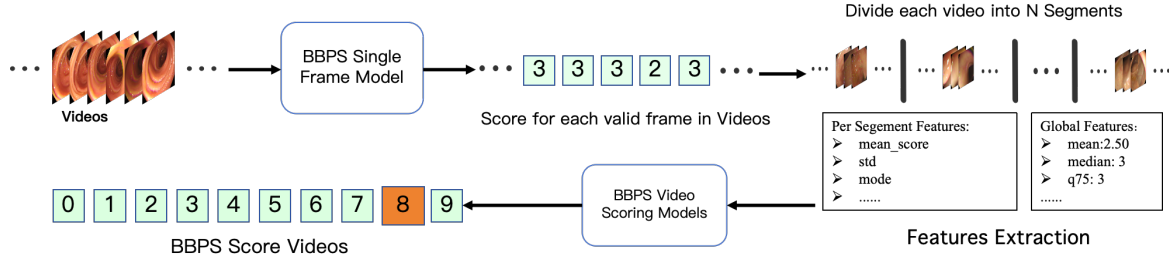


Fig. 2: Framework of Automated Bowel Preparation Scale Assessment

II. RELATED WORK

A. Artificial Intelligence (AI) Integration in Colonoscopy

The AI in colonoscopy procedures has been extensively explored to enhance various aspects of gastrointestinal diagnostics. Beyond the automation of the BBPS assessment, AI applications have been developed for polyp detection, classification, and segmentation. For example, the artificial intelligent endoscopist (AI-doscopist) is a deep learning model designed to localize colonic neoplasia during colonoscopy [2]. This system demonstrated a polyp-based sensitivity of 96.9%, successfully localizing 124 out of 128 polyps in a study involving 144 complete colonoscopies. Additionally, it achieved a specificity of 93.3% when evaluated on a per-image basis. Similarly, the Focus U-Net, a novel dual attention-gated CNN introduced by Yeung et al. [3], has been proposed for polyp segmentation. When evaluated across a combination of five public polyp datasets, the Focus U-Net achieved a mean DSC of 0.878 and a mean Intersection over Union (IoU) of 0.809, marking improvements of 14% and 15%, respectively, over previous state-of-the-art results. Recent works have leveraged deep learning-based methods to improve scene classification [4] and lesion detection [5]–[8] in endoscopy videos, demonstrating the potential of AI for enhancing gastrointestinal image analysis. Recent advancements in medical image analysis using hierarchical learning frameworks and active learning techniques have demonstrated the effectiveness of structured classification and efficient data annotation in improving diagnostic model performance [9], [10].

B. Automated AI Systems for Bowel Preparation Evaluation

Several AI-driven systems have been developed to automate and standardize evaluations. The Open-Source Automatic Bowel Preparation Scale (OSABPS) introduced by Cold et al. [11] is one such system capable of instantly assessing bowel preparation quality without human bias. Studies have shown that colonoscopies with an OSABPS score above a certain threshold should be considered for re-examination due to a lower polyp detection rate, indicating potential missed adenomas. Another notable development is the e-BBPS system, Yao et al. [12] developed a real-time automatic bowel preparation assessment tool. In a prospective study with 393 patients undergoing screening colonoscopy, the e-BBPS system effectively identified patients at high risk of missed adenomas

larger than 5 mm due to inadequate preparation. Similarly, another important advancement by Wang et al. [13] proposed a 3D deep learning system that utilizes spatiotemporal features from colonoscopy videos to assess bowel preparation. Their approach showed improved accuracy and consistency over 2D models. However, the system requires large annotated video datasets and is computationally intensive, which may limit real-time clinical deployment. Frame-level classification and temporal sequence aggregation strategies have been successfully applied to EGD video analysis for gastric location tracking [14], providing a technical foundation for endoscopic video-based quality or cleanliness assessment tasks.

C. Machine Learning Methods

As AI methodologies expand into medical diagnostics, combining deep learning approaches with traditional machine learning techniques through feature engineering has demonstrated additional improvements in predictive performance. Support Vector Machines (SVM) [15], have been widely applied due to their strong generalization capabilities, especially in scenarios with high-dimensional feature spaces. Random Forest algorithms [16], which construct multiple decision trees to reduce overfitting and increase stability. Gradient Boosting [17] sequentially builds decision trees by minimizing prediction errors and has been recognized for its excellent predictive accuracy, particularly in medical imaging and classification tasks. ElasticNet [18] combines the penalties of lasso (L1) and ridge (L2) regression, has proven effective in handling correlated features, preventing overfitting. Linear regression [19], despite its simplicity, remains fundamental due to its interpretability and ease of use, making it an essential baseline method for evaluating more complex algorithms.

III. METHOD

In this study, we introduce a comprehensive, multi-stage framework as shown in Figure 2, designed to automate the assessment of bowel cleanliness in colonoscopy videos using BBPS. The proposed framework integrates deep learning for frame-level classification with statistical feature engineering and traditional machine learning for video-level score prediction. Specifically, we first train image-based models to classify individual colonoscopy frames according to BBPS criteria. These models are subsequently applied to full-length videos to generate frame-wise cleanliness scores. Finally, a set of

handcrafted features is extracted from the sequence of frame scores and used to train regression models that produce a final BBPS score for each video. This pipeline enables efficient, objective, and reproducible evaluation of bowel preparation quality across large-scale colonoscopy datasets.

A. BBPS Single Frame Model

We use image dataset to train BBPS Single Frame Model. We leverage the models available in the PyTorch Image Models (timm) repository provided by Hugging Face [20]. In experiments, we utilize several topical image-classified architectures, including EfficientNet, Swin Transformer, Resnet, DenseNet, Vision Transformer (Vit) and MobileNet. These models are chosen for their balance among accuracy, number of parameters and computational efficiency. They have been pre-trained on large-scale image datasets, which provides a good starting point for transfer learning in our domain. The process begins with fine-tuning these models on our collected image dataset. In this step, we adjust the models' hyperparameters, such as learning rate, batch size, and weight decay, to optimize performance on our task, and we implement data augmentation techniques, such as random rotations, flips, and brightness adjustments, which are applied during training to improve the robustness and prevent overfitting. The training process uses a standard cross-entropy loss function, and optimization is performed using stochastic gradient descent (SGD) or Adam, depending on the specific model architecture.

In order to improve the accuracy of classification, we develop a new sequence model, shown in Figure 3. Instead of treating the BBPS scoring as a straightforward multi-class classification problem, we adopt a hierarchical approach with sequence binary classification models. In the first layer, we divide the dataset into two groups: one containing images with a class between 0 and 3 (valid images) and another containing images in class 4 (redundant images). This binary classification allowed the model to focus on differentiating between usable images and unrecognized cases. In the second layer, the images in classes 0-3 are further subdivided. The model distinguishes between images with classes of 0 and 1 (relatively cleaner) versus those with classes of 2 and 3 (relatively dirtier). In the third layer, we have two sub-models. The first sub-model divides images between class 0 (cleanest) and class 1 (cleaner), while the second sub-model divides images between class 2 (dirtier) and class 3 (dirtiest). Then, we combine these four binary models to become a big 5-class classification model. This hierarchical binary splitting strategy leads to improved performance, as it allows the model to focus on finer distinctions within a narrower range of scores. In our experiments, this approach resulted in an increase of classification accuracy by approximately 3-7% compared to a single-stage multi-class classifier.

The trained BBPS single-frame model is then applied to the video dataset. Since not all frames in the videos contain valid or relevant bowel images, the doctors first manually select segments containing valid bowel content from each video. We then apply our trained BBPS single-frame model

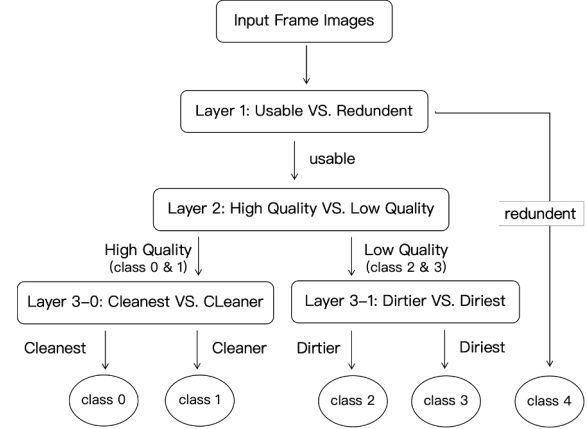


Fig. 3: BBPS Single Frame Model

to predict the bowel cleanliness scores for each valid frame. In conventional BBPS assessments, scores are assigned based on specific colon segments: ascending colon, transverse colon, and descending colon. Each segment receives a score from 0 to 3, determined by clinicians who evaluate selected video segments clearly identified by their anatomical locations. The final segment scores are typically averages of multiple scored frames within each region. The total BBPS for a video is the sum of these three segment scores, yielding a final score range between 0 and 9. However, instead of assigning scores based on anatomical location, our model predicts a cleanliness score for every valid frame independently without considering their anatomical positions within the colon, and frames classified as the redundant class 4 are discarded. After filtering out these invalid frames, we splice the remaining valid frames together to form continuous video segments suitable for subsequent scoring and analysis. In our BBPS study, the images in class 0 get 3 points, because they are the cleanest part of the images; the images in class 1 get 2 points; the images in class 2 get 1 point, and the images in class 3, which is the worst condition of colon get 0 point. Subsequently, these frame-level predictions are aggregated and analyzed collectively through machine learning techniques to generate a comprehensive video-level cleanliness score.

B. Feature Extraction

We extract features from the frame score data generated from the last step. Considering the input data for machine learning algorithms, including SVM, Random Forest, Gradient Boosting Trees, etc., must be structured numerical feature vectors of fixed length, each video is segmented uniformly, typically into 10 to 50 segments, to capture both local and global variations in quality throughout its duration. We extract three main categories of features per segment: the first category includes the proportion of high-quality frames (frames with scores 2 and 3), the proportion of low-quality frames (scores 0 and 1), and the mean quality score within the segment. Secondly, we compute the standard deviation of frame scores

as a measure of quality fluctuation and diversity, defined as the ratio of unique frame scores relative to the possible scoring range. Additionally, the most frequent score (mode) is determined, offering insight into the dominant quality level present. Thirdly, to capture temporal trends, we perform linear regression of quality scores against frame indices within each segment. From these regressions, we extract the slope (indicating increasing or decreasing trends in quality), the mean frame-to-frame change in quality, and the proportion of frames exhibiting improvements over their predecessors. In addition to segment-level descriptors, we compute several global summary features across the entire video, including the global mean score, the 25th and 75th percentiles of scores, and the global standard deviation of all frame scores. Figure 4 shows the feature extraction pipeline.

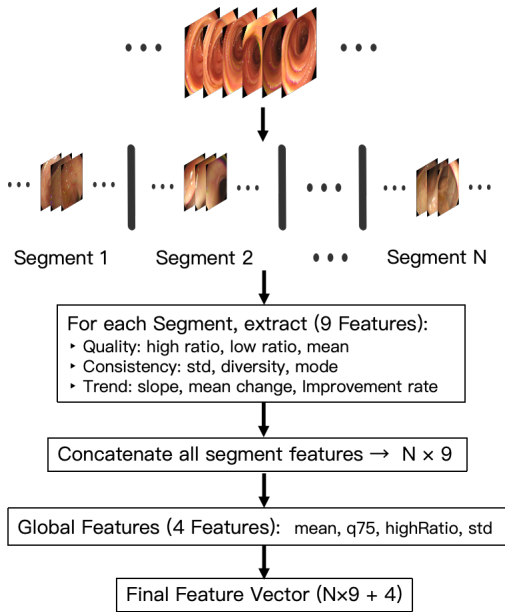


Fig. 4: Feature Extraction

Three distinct feature engineering strategies with different features are implemented to predict video ratings using machine learning algorithms. The first strategy, called “full feature analysis”, involves extracting nine features per video segment, including the proportion of high-quality frames, the proportion of low-quality frames, the segment’s mean score, the standard deviation, a diversity metric calculated from the ratio of unique quality scores, the mode, the trend, the average frame-to-frame changes, and the proportion of improvements between consecutive frames. Additionally, global descriptors—overall mean, 75th percentile score, overall high-quality ratio, and overall standard deviation are computed across the entire video. This comprehensive feature set is constructed by concatenating segment-level features and appending global features, forming the input for subsequent modeling. The second strategy, termed “focused feature analysis”, simplifies the feature set to three statistically prominent features per

segment and globally: the mode, mean score, and the 75th percentile. We will discuss why we choose these three features in the experiment section. This focuses subset aimed to preserve the most predictive and interpretable information about the video’s quality distribution while reducing complexity. The third strategy, the “mean-only” feature analysis, further simplifies feature extraction by exclusively using mean-based features at the segment level and globally.

C. Machine Learning Model

Five machine learning models are employed to predict bowel scores based on features extracted from segmented and aggregated video frame quality data. The models included SVR, Random Forest, Gradient Boosting, ElasticNet, and Linear Regression. Each model is implemented using the scikit-learn library and integrated into a unified training and evaluation pipeline. For each feature strategy: full (9 features), focused (mode, mean, q75), and mean-only, the feature matrix and corresponding labels are split into training and test sets. A grid search with five-fold cross-validation is applied to each algorithm to optimize hyperparameters. In SVR, both radial basis function (RBF) and linear kernels are explored with varying regularization and gamma parameters. The Random Forest model is tuned by varying the number of trees, tree depth, and minimum samples per split. The Gradient Boosting model similarly adjusts the number of estimators, learning rate, and tree depth. ElasticNet uses different combinations of alpha values and L1/L2 mixing ratios to control regularization. Linear Regression configurations include fitting the intercept and testing the effect of positive-only constraints. Model performance is assessed using standard regression metrics, including mean squared error (MSE), mean absolute error (MAE), coefficient of determination (R^2), and explained variance. For each model, predictions from cross-validation folds and the held-out test set are collected to evaluate both generalization and robustness.

IV. EXPERIMENTS

A. Dataset

The data used in this study is derived from real-world colonoscopy procedures, originally collected in [21]. It includes examinations from 256 individual patients, ensuring our samples are representative of genuine clinical scenarios. As detailed in Table I, we extract frame-level image data from 110 colonoscopy videos, which is primarily used for model training and validation. Table II summarizes the distribution of the video-level dataset, which is reserved for evaluating the system in realistic settings.

TABLE I: Image Dataset Distribution

BBPS Class	Training	Validation
0	1,117	315
1	1,103	126
2	953	366
3	1,025	247
4	1,633	1,248
Total	5,831	2,302

TABLE II: Video Dataset Distribution

Score	Count	Score	Count	Score	Count
1	3	4	15	7	30
2	6	5	20	8	13
3	7	6	48	9	4

The image data are categorized according to the BBPS into five classes ranging from 0 to 4. Images classified as 0 through 3 are considered valid for assessment, whereas images labeled as class 4 are redundant because the image is so poor that its cleanliness cannot be evaluated. In this system, class 0 represents the cleanest bowel, showing virtually no residual debris. Class 1 indicates a mostly clean bowel with only minor areas of dirtiness. Class 2 reflects a moderately dirty bowel, while class 3 denotes a very dirty bowel where the overall structure is still discernible, though with significant obstructions. These images are extracted from colonoscopy videos and undergo several preprocessing steps. Initially, we remove the irrelevant black borders around each colonoscopy frame by detecting and cropping out regions without intestinal visual information. After border removal, all images are normalized for brightness and contrast levels and are resized to standard dimensions, ensuring uniformity and facilitating consistent model input.

The second part of the data is the video dataset, which consists of 146 colonoscopy videos. As shown in Table II, the dataset is markedly imbalanced, with most videos clustered around intermediate BBPS scores, which reflects the clinical reality that both severely unprepared and perfectly clean bowels are infrequently encountered. The videos are annotated by experienced gastroenterologists, who marked the effective parts of each video, gave each segment a score, and also marked the location of each segment (ascending colon, transverse colon, descending colon). Gastroenterologists obtained the total Boston score of the video based on the score of each location. The score data from doctors serves as a ground truth for evaluating the performance of our AI system. Moreover, the videos are processed frame by frame, and predictions from the model are aggregated to form a continuous assessment that can be directly compared with the clinicians' scores.

B. Experimental Setup and Evaluation

All model training and inference experiments are conducted on a workstation equipped with four NVIDIA TITAN RTX GPUs, each with 24 GB of dedicated memory.

1) *Single frame model*: In the BBPS single frame model training experiments, we set some hyperparameters for all models: learning rate is 0.05, dropout is 0.3, batch size is 32 and epochs is 100. Each model is trained for a fixed number of epochs, and the best-performing model is selected based on validation set performance.

After the model is trained on the images, we incorporate Class Activation Mapping (CAM) techniques to verify that the models have learned the relevant features. Specifically, we use Grad-CAM (Gradient-weighted Class Activation Mapping) as implemented in the GitHub repository [22]. By generating heat maps over the input images, Grad-CAM provides visual

evidence of which regions in the colonoscopy images contribute most to the model's predictions. This step is crucial for validating that the model is not merely overfitting to irrelevant features but is instead focusing on areas that correspond to the clinically important parts of the colon. In well-prepared images class 0, the model should emphasize areas where the colon surface is clearly visible, while in poorly prepared images class 1-3, the model's attention should be directed to regions obscured by residual stool or opaque liquid. Figure 5 is a cam example in class 0 and 3.

We evaluate multiple backbone models using two strategies: a single-stage 5-class classification and a hierarchical sequence of binary classifiers. In the single-stage classification approach, each model directly distinguishes among five classes, including redundant class-4 frames. Additionally, to explore the impact of redundant frames on classification performance, we also evaluate a simplified 4-class version excluding images from class 4. The corresponding validation accuracies for both scenarios are presented in Table III. Considering the number of images in class 4 is much more than the sum of the number of images in class 0-3 and it is easier for models to recognize redundant images, it is reasonable that the models' performance in 5-class is much better than in 4-class.

As discussed before, we propose a 3-layer hierarchical classification strategy to enhance classification accuracy, involving a sequence of binary classifiers. Specifically, the first binary classifier separates valid (classes 0-3) from redundant (class 4) frames. The second layer distinguishes relatively cleaner frames (classes 0 and 1) from dirtier frames (classes 2 and 3). The third layer comprises two binary classifiers that further discriminate between classes 0 versus 1, and classes 2 versus 3, respectively. Table III summarizes the individual validation accuracies for these binary models. Within this hierarchical approach, classifiers distinguishing between classes 2 and 3 typically yield the lowest performance, with accuracies around 80%. Conversely, classifiers distinguishing between valid and redundant frames, cleaner versus dirtier frames, and class 0 versus 1 consistently achieve high accuracy, generally above 93%. By integrating these binary classifiers, we form an overall 5-class classifier, whose final validation accuracy is also shown in Table III.

Compared to the single-stage 5-class models, the hierar-

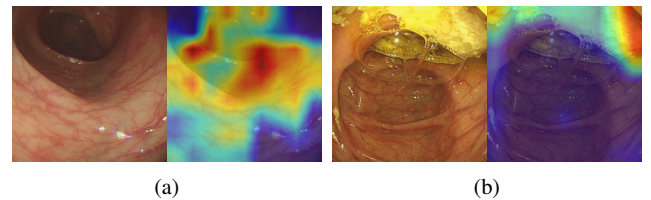


Fig. 5: Grad-CAM visualization on BBPS classes. (a) Class 0 (clean bowel), highlighting clear mucosal areas without obstructions. (b) Class 3 (poorly prepared bowel), emphasizing regions obscured by residual stool and liquid, which significantly impede mucosal visibility.

TABLE III: Comparison of Model Performance and Parameters

Model	5-class model Best acc@1(%)	3-layer sequence binary models(%)	class 0-3 vs 4(%)	class 0,1 vs 2,3(%)	class 0 vs 1(%)	class 2 vs 3(%)	Params (M)
efficientnet_b3a	87.80	89.36	97.35	94.97	93.65	81.89	18.9
swin_base_patch4_window7	88.54	89.01	96.22	95.25	95.01	84.50	87.8
densenet201.tv	86.41	86.79	95.74	95.06	93.19	79.05	20.0
mobilenetv3_large_100	86.98	87.79	95.96	94.40	93.42	81.89	5.5
vit_base_patch32	81.34	88.36	96.17	94.78	92.74	83.52	104.3
resnet101	81.91	87.14	95.43	93.64	93.42	81.40	44.5

TABLE IV: BBPS Video Scoring Machine Learning Models Hyperparameter Settings

Model	Hyperparameters and values explored
Support Vector Regression (SVR)	Kernel={rbf, linear}, C={0.01,0.1,1,10,100}, γ ={scale,auto}, ϵ ={0.01,0.1}
Random Forest	$n_estimators$ ={100,300,500}, max_depth ={None,10,30,50}, $min_samples_split$ ={2,10}, $min_samples_leaf$ ={1,2,4}
Gradient Boosting	$n_estimators$ ={100,200,500}, $learning_rate$ ={0.01,0.05,0.1}, max_depth ={3,9}, $min_samples_split$ ={2,10}
ElasticNet	$alpha$ ={0.01,0.1,1.0}, $l1_ratio$ ={0.1,0.3,0.7}, $selection$ ={cyclic,random}
Linear Regression	$fit_intercept$ ={True}

chical sequence of binary classifiers consistently demonstrates superior performance across all backbone models, achieving accuracy improvements ranging from approximately 0.3% to 7%. For example, the EfficientNet-B3 model's accuracy increases from 87.80% to 89.36%, an improvement of 1.56%. The ViT-Base model achieves the greatest improvement, rising from 81.34% to 88.36%, reflecting a substantial gain of 7.02%. These results confirm the effectiveness of the hierarchical binary classification strategy for improving BBPS assessment accuracy.

2) *Video-level model*: We extend our approach to colonoscopy videos. As we discussed before, the first step is feature extraction. Our method intentionally omits positional information, and we combine all usable frames together as a valid video. We then divide every video into identical segments, and extract 9 features in each segment, Figure 6 shows an example of segment analysis. To evaluate the relative contribution of different categories of extracted features to video-level BBPS score prediction, we conduct a feature type importance analysis. For each feature type, we calculate its importance by computing the average absolute Pearson correlation coefficient between all features of that type and the corresponding ground-truth video scores across the dataset. In all the BBPS video scoring experiments, the Q75, mean, mode, and high quality are the four most important features, and considering the high quality (score in 2 and 3) is hugely influenced by datasets, we decide to choose Q75, mean, and mode as the main features to do "focused feature analysis". An example of feature importance analysis is shown on Figure 7. Although Q75 emerges as the most important feature in feature analysis, we ultimately select the mean frame score as the sole feature for the third strategy, because in the context of frame values restricted to only four discrete categories (0, 1, 2, 3) and its value can be misleading for overall video-level scoring, the mean provides a stable and interpretable summary of overall video quality. With these extracted features, we experiment with five classical machine learning models discussed before to predict the final video-level BBPS score. Each model is trained and evaluated using five-fold cross-

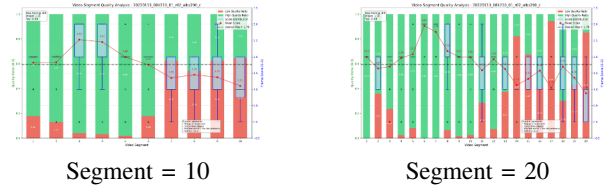


Fig. 6: Feature extraction and analysis with different segments. For each segment, the proportion of high- and low-quality frames is visualized using stacked bar plots, while the distribution of frame-level quality scores is shown as a boxplot. The mean score for each segment is indicated by a red line, and the overall mean and expert rating are marked for reference.

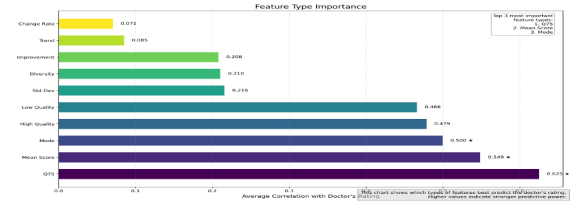


Fig. 7: Feature Importance Analysis

validation to ensure robustness and avoid overfitting, measured by MSE, for each hyperparameter combination. The optimal set of hyperparameters for each algorithm is selected based on the lowest average cross-validation MSE. The selected hyperparameters and settings for each model are shown in Table IV.

To comprehensively evaluate the performance of our automated BBPS video scoring pipeline, we design a series of controlled experiments focusing on five main factors: 1) The number of segmentation, 2) Machine learning model selection, 3) Frame score generation approach (single 5-class model or sequenced 2-class models), 4) Feature engineering strategies, 5) The basic BBPS single frame model selection.

To minimize the influence of confounding variables, all comparisons in this section employ EfficientNet-b3 as the

TABLE V: Video Score Performance Comparison (Based on efficientnetb3)

Segment	BBPS Single Frame Model	Feature Strategy	Best ML model	Accuracy (%)	
				Exact Match	Within-1
10	sequence 2-class	Full Focused Mean-only	Random Forest	36.7	93.3
			Gradient Boosting	43.3	90.0
			Random Forest	40.0	86.7
	single 5-class	Full Focused Mean-only	Random Forest	40.0	93.3
20	sequence 2-class	Full Focused Mean-only	Random Forest	53.3	93.3
			Random Forest	50.0	93.3
			Gradient Boosting		
	single 5-class	Full Focused Mean-only	Gradient Boosting	43.3	93.3
			Gradient Boosting	43.3	93.3
			Gradient Boosting	33.3	93.3
30	sequence 2-class	Full Focused Mean-only	SVR	40.0	93.3
			SVR	40.0	90.0
			SVR	40.0	93.3
	single 5-class	Full Focused Mean-only	Gradient Boosting	43.3	93.3
			Gradient Boosting	43.3	90.0
			Gradient Boosting	43.3	83.3
40	sequence 2-class	Full Focused Mean-only	Random Forest	40.0	96.7
			Random Forest	40.0	93.3
			Random Forest	40.0	90.0
	single 5-class	Full Focused Mean-only	Random Forest	36.7	90.0
			Random Forest	43.3	90.0
			Random Forest	43.3	90.0
50	sequence 2-class	Full Focused Mean-only	SVR	36.7	93.3
			SVR	33.3	90.0
			SVR	33.3	90.0
	single 5-class	Full Focused Mean-only	Gradient Boosting	43.3	86.7
			Gradient Boosting	46.7	86.7
			SVR	33.3	86.7

backbone for the frame-level BBPS scoring model, thereby reducing the number of experimental factors to four. In addition to automatically selecting the best BBPS Video Scoring machine model, we introduce a comprehensive ranking system that evaluates performance across multiple key metrics: R^2 , Explained Variance, MSE, and MAE. For each metric, all candidate models are ranked from best to worst according to whether a higher or lower value is desirable. These individual ranks are then summed to produce a total score for each model. The model achieving the lowest aggregate rank is ultimately chosen as the best performer, ensuring a balanced and fair assessment that avoids biases from the different scales of the evaluation metrics.

The performance of video score experiments is in Table V. All the results are based on the best machine learning model in each experiment. To enable robust and clinically meaningful evaluation, we report both Exact Match Accuracy and Within-1 Accuracy for all experimental configurations. The inclusion of the Within-1 metric reflects the inherent variability and subjectivity in clinical BBPS scoring, where minor discrepancies of one point are generally considered acceptable among expert raters.

The results reveal several noteworthy trends. First, the number of video segments, ranging from 10 to 50, exerts only a modest influence on predictive accuracy. Across all

segmentations, both exact match and within-1 accuracy remain relatively stable. This finding suggests that our feature extraction approach robustly captures relevant information regardless of the granularity of video partitioning, supporting its generalizability across different video segmentation schemes.

Second, among the three feature engineering strategies, full, focused, and mean-only, the differences in predictive accuracy are marginal. In almost all segment and model combinations, the mean-only strategy achieves performance nearly equivalent to that of more complex feature sets, or even better, which frequently surpasses 90%. For instance, with 10 segments and the 5-class single-frame model, the mean-only feature set yields a within-1 accuracy of 93.3%, virtually indistinguishable from the comprehensive and focused sets. This convergence of results strongly indicates that the mean frame score is the single most informative predictor of video-level BBPS. Practically, this insight enables substantial simplification of the prediction pipeline, as accurate video-level scoring can be achieved with only the mean of frame-level predictions.

Third, while the adoption of a hierarchical sequence of binary (2-class) classifiers for frame-level BBPS assignment yields modest gains in frame-level accuracy (about 1–3% improvement over direct 5-class prediction), these benefits do not consistently translate into enhanced video-level scoring performance. Across all segmentation and feature strategies,

the accuracy of the 2-class sequence approach is similar to that of the standard 5-class model when evaluated at the video level. For example, the highest exact match accuracy observed is 53.3% (10 segments, 5-class, focused features, Random Forest), which slightly exceeds the best sequence model result (46.7%, 50 segments, 2-class, focused features, Gradient Boosting).

In summary, overall within-1 accuracy across all models and configurations is consistently high, frequently reaching or exceeding 90%. This high level of agreement with expert ratings underscores the clinical applicability of the proposed pipeline, particularly given the known subjectivity and inter-rater variability of manual BBPS scoring.

V. CONCLUSION AND FUTURE WORK

We present a comprehensive, data-driven framework for fully automated BBPS video scoring in colonoscopy. By introducing a novel hierarchical sequence binary classification strategy for frame-level scoring and systematically benchmarking a wide range of feature engineering and machine learning approaches at the video level. Notably, the hierarchical sequence of binary classifiers improves frame-level accuracy by approximately 0.3%-7% compared with direct multi-class prediction, highlighting its effectiveness in handling the inherent difficulty of fine-grained cleanliness categories. Our results further demonstrate that the mean of predicted frame scores alone can reliably approximate expert video-level ratings, with "within-1" accuracy consistently exceeding 90% across almost all experimental configurations.

This study has several limitations. First, the video dataset is relatively small, which restricts model robustness and generalizability. Second, the distribution of BBPS scores across videos is imbalanced, rare classes are underrepresented, yet this imbalance also reflects the reality of clinical practice, where extreme cleanliness or severe soiling is less common. Third, physician-assigned BBPS labels are inherently subjective, and inter-rater variability may introduce noise that affects both training and evaluation. Finally, although our system performs well in retrospective testing, real-time deployment still requires computational optimization and regulatory validation. Future work will expand the dataset through multi-center collaborations, address imbalance with augmentation or reweighting, and incorporate consensus or multi-rater annotation strategies to mitigate subjectivity.

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